Comon Terms And Concept

1 pathogene - An organism that causes a teases in plant (in a host)
2 phyto-pathogenes - These are organisms that can cause a disease in a host

3 Abiotic deseases - are caused by non-living agents such as temp, environment
4 Biotic diseases -

5 parastisms - Relationship ~~between~~ two living organism where one organism benefit at the expense ~~of~~ the
6 Epiphytes - plants that grows on another plants for support
7 Saprophytes - These are organisms that live on the dead and gets it's nourishment from the decaying organic matter (fungus
8 pathogeneecity - is the abiility of an organism to cause a disease ~~other~~ as high pathogeneecity or low.
9 Acquired resistance : The resistance by the plant from diseases through treatment by reproduction of certain Chemicals.
10 parasite: An organism that live upon another organism to obtain food.
11 Biotrophs ↗ Plant pathogenic fungus that requries a living cell and drive its food from there
12 Hemibiotroph - plant pathogenic fungi that would start by living on the host and later kill the host, then continue to survive on the dead.

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13 Necrotrophs
- These are organisms that kills the host and survive on the dead cells.
14 Pathogenesis This are process that an infection agent will go through when it comes in Contact with Suseptible host
Susceptible

15 Infection - Invasion / initiation and establish of a parasite within a host plant.

16 Inocium A potion of a pathogene that is released into the host plant to Cause a disease.

17 Invansion - The penetration and spreading of a pathogene.

18 Colonisation - is the growth of a pathogen into a plant.

19 Symptons ; are external and Internal reaction or alteration of a plant as a result of the disease

20 Incubation period ; This the perod of time between the penetration of a pathogene into a host and a first appearance of Symptoms.

21 Disease Cycle : Those are a Chain of events that leads to disease development

22 Disease Syndrome : Are a set of varying Symptom Characterizing a disease

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23 Alternative host: These are plants that are not related to the main host of the pathogens but can be used to complete life cycle

24 Collateral host: Is where a plant is related to the main host

25 Predisposition these are effects of 1 or more environmental factors that makes a plant more vulnerable to attacks.

26 Symbiosis - This is the mutual beneficial relationship were organisms involved benefits

27 Mutualisms - mutual and (involves) a longer period

28 Antagonism The counter reaction of between a (groups) of organisms.

29 Mutation: This is the ~~abstract~~ change in appearance of a plant as a result of changes in gene composition

30 Crop damage: Any reduction in quantity and quality of the field

31 Deficiency: An abnormality or disease caused by lack of abnormal levels of essential nutrients.

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CONCEPT OF DISEASE IN PLANT

1 Unlike in human, plants do not feel pain or discomfort it is unknown
- A plant is considered normal if it is able to carry out physiological functions
* The meristematic should be able to divide and differentiate
* A plant must be able to carry out photosynthesis
* must be able to produce seeds
- If a plant is unable to do all these function it is said it has a disease

Plant treatment

1 observation of definite clear symptoms on a plant which give indication of the state of a plant. infected part of a plant give an indication of a type of disease
- Diseases are classified according to parameters
1. Disease symptoms,
2. Infected organ
3. Infected plant type
4. the type of phytopathogene.

??? Disease effects the plants that are in the in the same family forexample diseases in tomatoes effects only the plants in the family of Solonaceae which belong to the same family tomato mosaic

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- Indentification of phytopathogens is important because it helps to determine the disease cause, potential disease complications and offer possible control methods.

How pathogenic microorganism cause diseases in plants.

1. ~~Disease~~ Disturbing the metabolism of plants cells through enzyme, toxins, growth regulators, and other substances they secrete and absorbing food stuff from the host cells for their own ~~growth~~ use.
2. Some pathogen may also cause diseases by multiply in the xylem and phloem vessels of plants thereby blocking the upward transportation of water, downward movement of sugars, respectively through tissues.

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Types of Plant Diseases and Basic Procedures in The

Diseases

The plant disease diagnosis depend on the exact determination of disease cause.

1 The most comon method of diagnosing plant disease is by grouping them according to the symptom they cause.

2 Generally they are two plant disease causes: The pathogene and/or environment factors. "following are methods used"

i Do they cause root cause, leaf spots, blights, rot
ii the plant that is affect
iii The type of plant affected

1. Infectious Diseases / Biotic
- Disease caused by fungu
- " bacteria
- " parasitic high plants and green alga
- " virus and viroids
- " nematod

2 Non-Infectious Disease / Abiotic
- Diseas caused by too low or too high temp
- " lack or excess of soil moisture
- "

Two types of disease diagnosis:
1 Observation
2 laboratory.

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1.1.1 Disease Caused By parasitic higher plants

- Disease is caused due to growing on it and attaches itself on a plant they take all the required nutrients without benefit from it
- This cause abnormal weakness and others disease can be developed. It attaches on the surface of the host plant examples (epiphytes), dodda, witchweed and broomrape.

1.1.2 Diseases Caused by Nematodes

1.1 COMON FUNGAL DISEASES AND THEIR SYMPTOMS

- fungi are Eukaryotes i.e their cells contain membrane-bound organelle and clearly defined nuclei.
- They are small (microscopic)
- Some fungi known as obligate parasite or biotrophs, can grow and multiply only by remaining in association with their host plant in their entire life.
- Others are non-obligate parasite, require a host plant for a part of their life but can complete their life cycle on the dead organic matter.
- Some live on the living others on the dead matter

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1.1.2 Characteristics of plant pathogenic fungi

1 Morphology

⇒ Most of fungi have filamentous vegetative body called mycelium. The mycelium branches in all directions

- Individual branches are called hyphae

ii Reproduction

- fungi reproduce chiefly by means of spores. Spores are reproductive organs which consist of 1 or more cells which maybe formed as asexually or Sexually.
- Those that reproduce asexually, they reproduce in the sac called sporangium.
- Sexually it occurs in groups of fungi those that undergo sexual reproduction are called Zygomycetes, two cells of similar size and appearance unite and produce a zygote called Zygospore.

iii Dissemination

- Zoospores can move few millimeters (with flagellum tails)
- fungi dissemination can be done by water, animals, birds as spores

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Signs And Symptoms Caused By fungi on plants

- In most cases, fungi cause local or general Symptoms on their hosts. As such Symptoms may occur separately or Concurrently or may follow one another.
- In general, fungi cause local or general necrosis of plant tissues, and they often cause reduced growth (stunted growth or hypotrophy) of plants ~~tissues~~ organs or entire plants. A few fungi cause excessive growth of infected plants or plant parts.
- Symptoms of fungal diseases are leaf spots, blight, scab, rot, damping-off, diebacks and canker
- Symptoms associated with hyperplasia include galls, warts, and leaf curls.
- fungus affecting plants may produce growth or structures on the plant, stems, or leaves such as masses of mycelium

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- Examples of Symptoms Caused by fungi

1 leaf spots

2 leaf blight: This causes general and extremly rapid browning death of leaves, branches twings, and flora organs.

3 Canker: localised necrotic lesion on a stem or fleshly organs, often sunken

4 Die-back:

5 Root rot: Disintegration or decay of part of the root system of the plant

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tubers, e.t.c

7 Soft rots and dry rots: maceration and disintegration of fruits, roots, bulbs

8 Damping off: Rapid and Sudden Collapse of very young Seedlings.

9 Basal stem rot: Disintegration of the lower part of the stem.

10 Galls: Enlarged portions of plants organs (stems, leaves, ~~bot~~ blossom, roots)

11 Warts: Wartlike protuberances on tubers and stems

12 Witches-brooms: profuse, upward branching of twigs

13 Leaf Curls: Distortion, thickening, and curling of leaves.

14 Wilt: Generalized loss of turgidity and dropping of leaves and shoots.

15 Rust: many small lesions on the leaves or stems, usually of a rusty color.

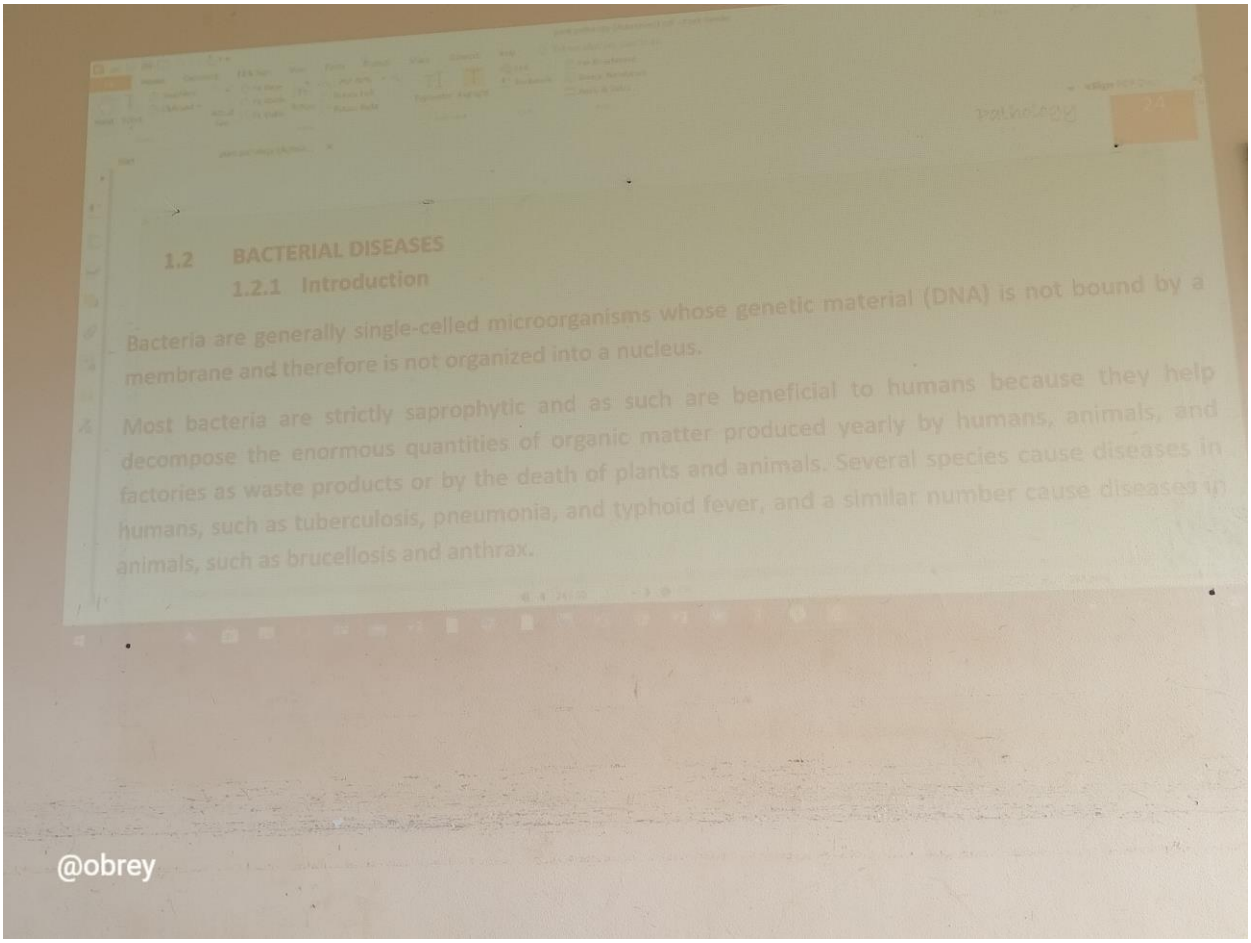
16 Smut: Seed or a gall filled with the mycelium or black spores of the Smut fungi

17 Mildew: Areas on leaves, Stems, blossom

Read further on PCR principle

fungal diseases has the majority of disease because fungal diseases can be quickly spread by wind, water

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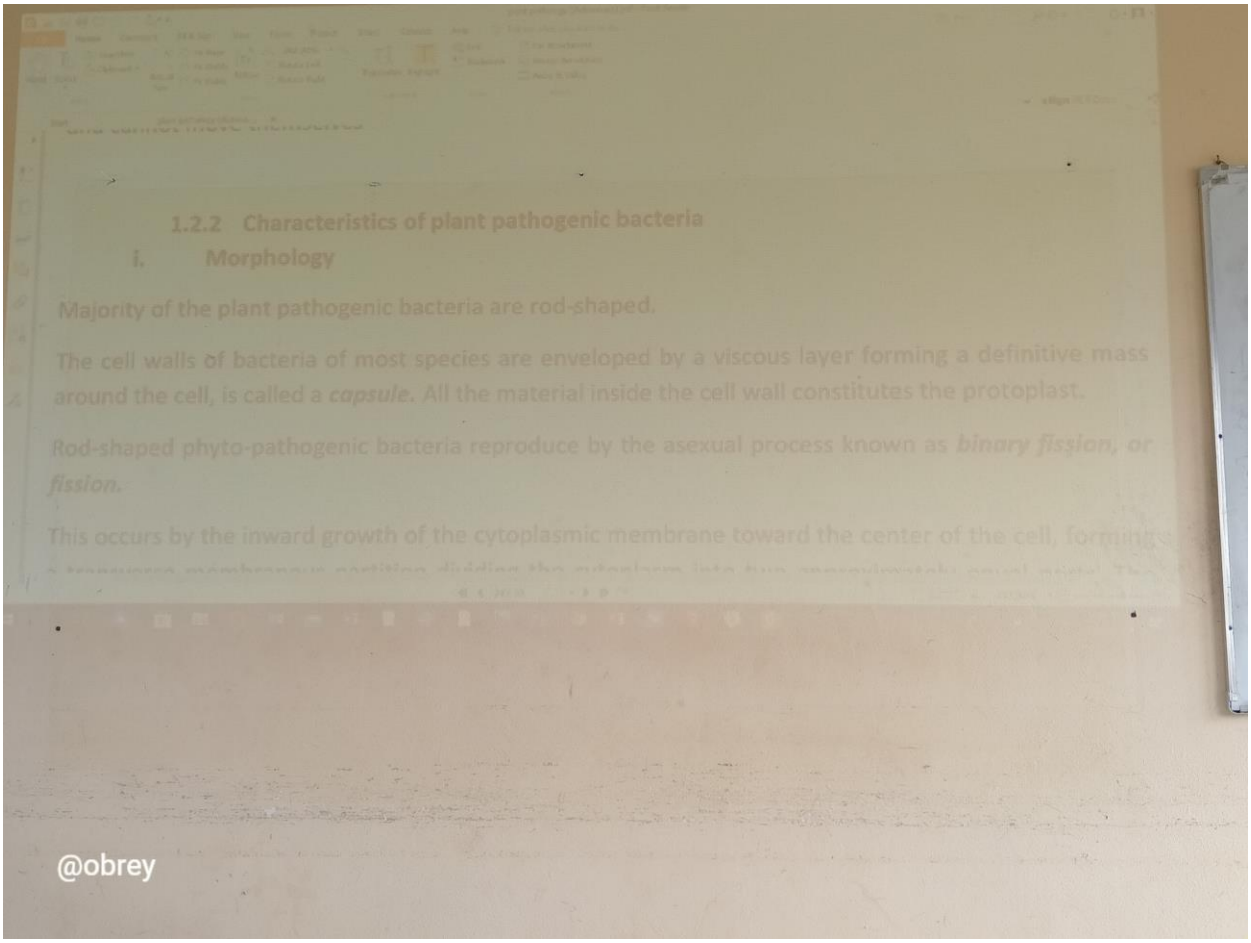
1.2 BACTERIAL DISEASES

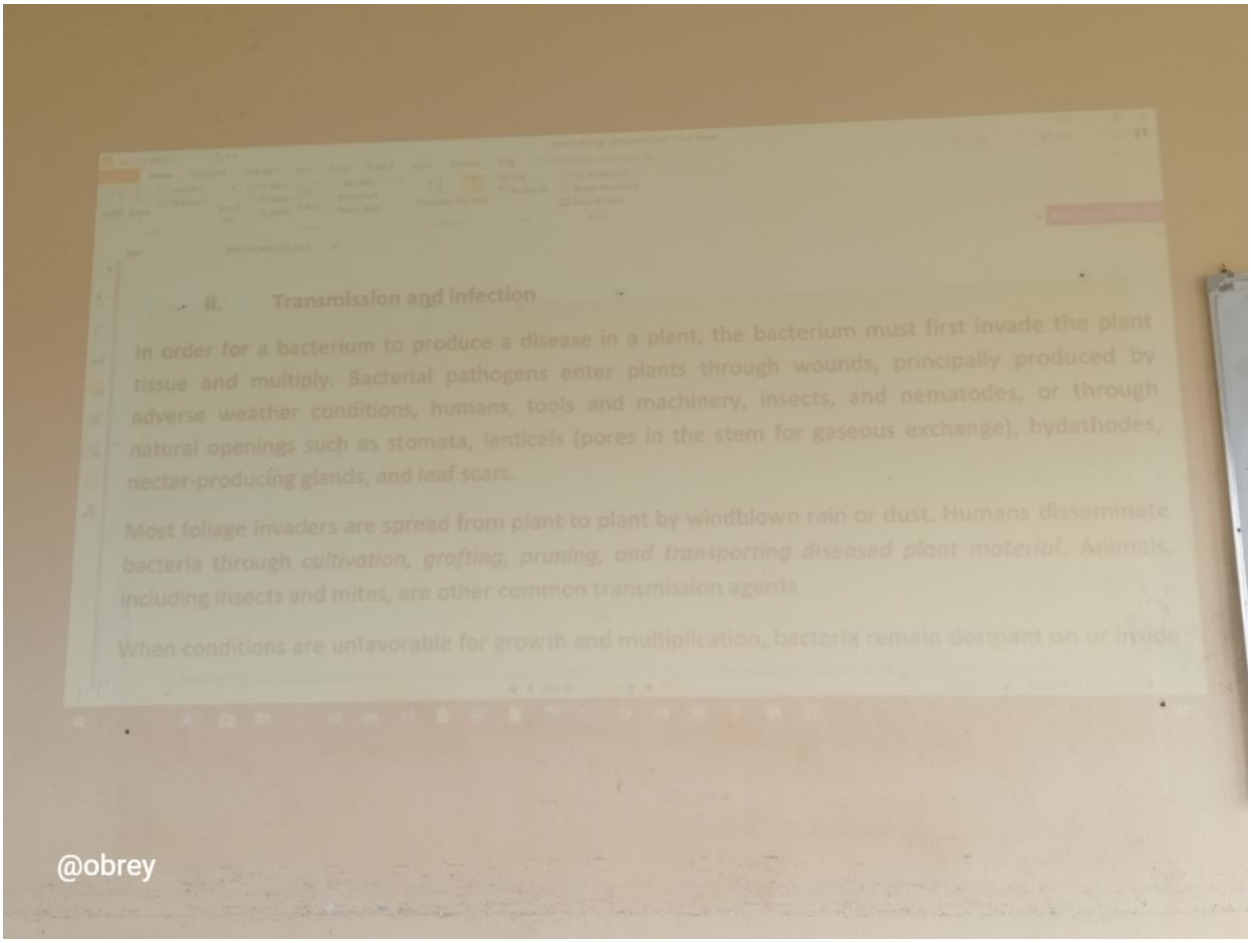
1.2.1 Introduction

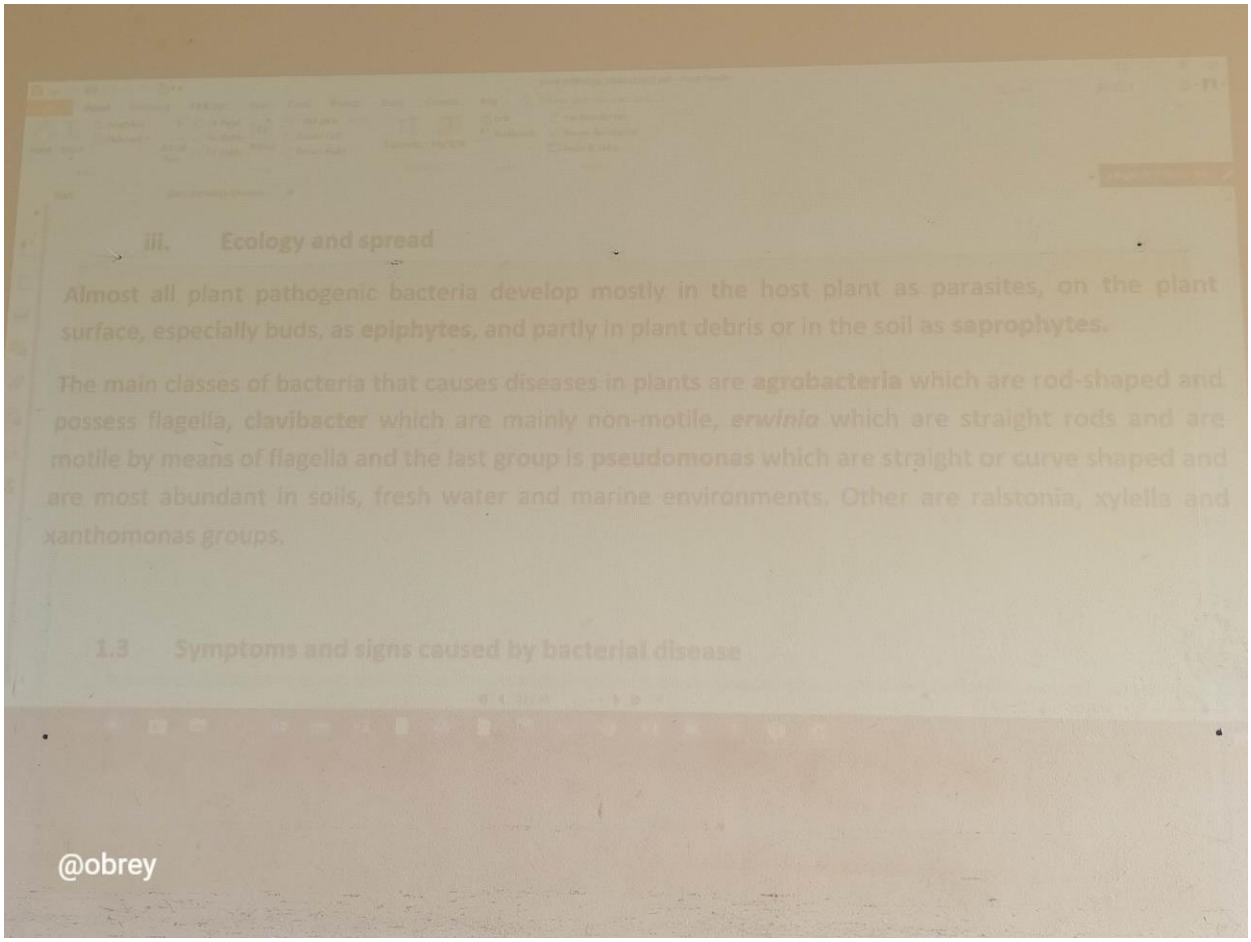
Bacteria are generally single-celled microorganisms whose genetic material (DNA) is not bound by a membrane and therefore is not organized into a nucleus.

Most bacteria are strictly saprophytic and as such are beneficial to humans because they help decompose the enormous quantities of organic matter produced yearly by humans, animals, and factories as waste products or by the death of plants and animals. Several species cause diseases in humans, such as tuberculosis, pneumonia, and typhoid fever, and a similar number cause diseases in animals, such as brucellosis and anthrax.

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1.2.2 Characteristics of plant pathogenic bacteria
i. Morphology
Majority of the plant pathogenic bacteria are rod-shaped.
The cell walls of bacteria of most species are enveloped by a viscous layer forming a definitive mass around the cell, is called a capsule. All the material inside the cell wall constitutes the protoplast.
Rod-shaped phyto-pathogenic bacteria reproduce by the asexual process known as binary fission, or fission.
This occurs by the inward growth of the cytoplasmic membrane toward the center of the cell, forming
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ii. Transmission and infection
In order for a bacterium to produce a disease in a plant, the bacterium must first invade the plant tissue and multiply. Bacterial pathogens enter plants through wounds, principally produced by adverse weather conditions, humans, tools and machinery, insects, and nematodes, or through natural openings such as stomata, lenticels (pores in the stem for gaseous exchange), hydathodes, nectar-producing glands, and leaf scars.
Most foliage invaders are spread from plant to plant by windblown rain or dust. Humans disseminate bacteria through cultivation, grafting, pruning, and transporting diseased plant material. Animals, including insects and mites, are other common transmission agents
When conditions are unfavorable for growth and multiplication, bacteria remain dormant on or inside
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iii. Ecology and spread
Almost all plant pathogenic bacteria develop mostly in the host plant as parasites, on the plant surface, especially buds, as epiphytes, and partly in plant debris or in the soil as saprophytes.
The main classes of bacteria that causes diseases in plants are agrobacteria which are rod-shaped and possess flagella, clavibacter which are mainly non-motile, erwinia which are straight rods and are motile by means of flagella and the last group is pseudomonas which are straight or curve shaped and are most abundant in soils, fresh water and marine environments. Other are ralstonia, xylella and xanthomonas groups.
1.3 Symptoms and signs caused by bacterial disease
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BACTERIAL DISEASES

- Bacteria are generally easyle-called microgonisms.
- Most bacterial are saprophitic and help human beings to decompose huge organic matter. (animals, waste product, plants)
- Several species of cause disease to humans
- Bacterial are in different Shapes and Sizes Some are rode Shaped, Spherical, Spiral or filamentous (threadlike)

Characteristics of plant pathogenic Bacteria

1 Morphology

- Majority are rod Shaped
- Most cell walls of bacteria are enveloped by a viscous layer forming a definite mass around the cell, called Capsule.
- All the material inside the cell wall Constitutes the protoplasm fission.
- Bacteria reproduce at a very rapid rate.
- Under favourable Conditions, bacteria may divide every 30 to 50 minutes, one bacteria becoming 2, 2 becoming 4, 4 becoming 8 over One million Can be formed one day.

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Transmission And Infection

- In order for a bacteria to reproduce a disease in a plant, the bacterium must first invade the plant tissue and multiply.
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- Bacteria desseminate from plant to plant by ~~bacteriocts~~ blindblown, dust rain.
- Dessimmmate through Cultivation, grafting, prunning and transporting deseased plant material. Insects and mites are effective transmitting agents
- Bacteria transmission are influenced greatly by temp and moisture. often, a different of only a few degree in temp determines weather a bacteria disease will develop. In most cases, moistures as a water film as a water on plant surfaces is essential for establishing an infection

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Ecology And Spread

- Almost all plant pathogenic bacteria develop mostly in the host plant as parasites, on the plant surface, especially buds, as epiphytes, and partly in plant debris or in soil saprophytes.
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Symptoms And Signs Caused By Bacteria Disease

Bacteria diseases can be grouped into four broad categories based on the extent of damage to a plant tissue and the symptom the cause, which include

1 Vascular wilt

This result from the bacteria invasion of the plant's vascular system. The subsequent multiplication and blockage prevents movement (translocation) of water and nutrients through the xylem of the host plant. Wilting, drooping or death

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occur; examples includes Rust of sweet Corn, tobacco, tomato and Cucurbits (squash, pumpkin and cucumber).

Necrosis

2 Necrosis

- pathogenes cause necrosis by secreting poison (toxin). Symptoms include spots, Stem blights or Cankers.

3 Soft rot

- Soft rot are caused by pathogenes that Secretes enzymes capable of decomposing cell wall structure, thereby destroying the tissue of the plant tissue. the plant tissue becomes macerated (soft and watery). This usually occur in fresh vegetables Such as potatoes, Carrot, eggplant Squash and tomato.

4 Tumors

- Tumor disease are caused by that stimulate uncontrolled multiplication of plant cells resulting in the formation of abnormally large structures.

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Examples of Bacterial Diseases

1 Bacterial spots and blight:

- These appear of spots of varies sizes on leave, stem, blossom and fruits. In some bacteria diseases the spots continue to advance rapidly and then the disease is called blights.

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Management And Control

Control of plant Diseases

- Bacterial diseases are very difficult to Control.
- A Combination of measure can be used to Control

1. **Chemical Control:** Involves the use of Chemicals like fungicide, bactricides to Control plants disease
2. **Genetic Control:** involve breading disease resistant plant which creates new Cultivar or varieties that peform better.
3. **Cultral practicess:** this involves the use of various practice that aim at at avoiding the provision Conducive environment for their Survival and multiplication of the of the pathogene. for example Crop rotation to disrupt year to year past cycles
4. **Regulatory Controls:** these are based on exclussion of pathogenic organism, highly Susceptible host plants, and alternative host plants from a particular area.
5. **Biological control:** this essentially involves using nature to prevent or treat disease
6. **Sanitation practice:** aiming at reducing the inoculum in a fieed by removing and burning
7. Adjusting fertilizing and watering: this is done so that the plants are not extremly Succulent during the period of infection may also reduce the necdence of disease

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8 Crop rotation

9 Seed treatment: Hot water at about 50°c (122°F) is also effective against management of bacterial disease in carrots, eggplant, pepper and tomatoes.

10 Biosecurity measure: also important to the transfer of disease pathogenes from an infected area to another.

11 Isolation: as it is usually called, purification of viruses is a very effect methods of managing viral pathogenes.

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